



Report ID: production-cic-ids2017

Source: CIC-IDS2017 packet windows · ABSTAINED · Generated 2026-09-17T14:17:02.637Z

VERIFIED AUDIT RECORD

EXECUTIVE SYNTHESIS

Executive Summary

High-level assessment of current network threat signal and forward multi-step trajectory.

CURRENT NETWORK ASSESSMENT

BENIGN TRAFFIC
BASELINE

Passive traffic conforms to baseline statistical operating bounds.

FORECAST HORIZON

T+1 through T+5 (300
Seconds)

Continuous state dynamic rollout over 5 discrete 60s tumbling windows.

PRIMARY SIGNAL

mean_iat

Top feature driver identified via counterfactual perturbation.

FORECAST OUTCOME

RECONNAISSANCE →
STABLE_BENIGN

Rollout trajectory across lookahead windows. Current threat signal does not imply guaranteed future escalation.

SEMANTIC DISTINCTION NOTICE: **Current Network Assessment** reflects empirical telemetry anomalies detected in observed wire traffic at T0. **Forecast Outcome** reflects multi-step forward neural state rollouts. An observed threat signal at T0 indicates active anomalous pressure but does not represent a confirmed or inevitable future breach; trajectory may stabilize, localize, or decay as rollout deepens.

Input Telemetry

Source File: CIC-IDS2017 packet windows

Format: PARQUET (Passive wire capture)

Capture Duration: 120s (2 windows)

Observed Volume: 12 pkts / 3,120 flows

Current Network State (S_t)

State: 45-Dim Continuous State
Formulation: Vector

Window Stride: 60s discrete tumbling slices

Throughput in Window: 12 pkts

Endpoints: 12 source · 4 destination hosts

Multi-Horizon Forecast Rollout

Forward-looking neural state rollout and uncalibrated transition dynamics across lookahead horizons.

HORIZON	STEP	SCORE	COMPOUNDING RISK	PREDICTED STAGE	SUPPORTING SIGNALS
T+1	(+60s)	N/A	N/A	RECONNAISSANCE	Forecast withheld: Insufficient continuous temporal history (< 8 windows).
T+2	(+120s)	N/A	N/A	RECONNAISSANCE	Forecast withheld: Insufficient continuous temporal history (< 8 windows).
T+3	(+180s)	N/A	N/A	RECONNAISSANCE	Forecast withheld: Insufficient continuous temporal history (< 8 windows).
T+4	(+240s)	N/A	N/A	RECONNAISSANCE	Forecast withheld: Insufficient continuous temporal history (< 8 windows).
T+5	(+300s)	N/A	N/A	RECONNAISSANCE	Forecast withheld: Insufficient continuous temporal history (< 8 windows).

Methodological Disclosure: Step Scores and Compounding Risk represent uncalibrated model transition scores and latent state activations, not empirical or actuarial probabilities of attack occurrence. Forward projections reflect statistical dynamics learned from training distributions.

Predicted Attack Progression

Sequential behavioral evolution across the kill-chain derived from transition matrix dynamics.

Temporal Horizon vs. Behavioral Stage: Lookahead horizons (T+1 to T+5) represent discrete 60-second forward temporal windows (+60s to +300s). Attack stages (RECONNAISSANCE → STABLE_BENIGN) describe discrete MITRE ATT&CK tactic classifications mapped to those windows via transition matrix dynamics, not continuous elapsed duration.

STAGE 1 (+60s) STATE_PERSISTENCE RECONNAISSANCE Transition Signal: 88%	STAGE 2 (+120s) STATE_PERSISTENCE PORT_SCAN Transition Signal: 79%	STAGE 3 (+180s) DOWNSTREAM_PROGRESSION EXPLOITATION Transition Signal: 68%	STAGE 4 (+300s) DOWNSTREAM_PROGRESSION LATERAL_MOVEMENT Transition Signal: 54%
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Contextual MITRE ATT&CK Interpretation

Contextual behavioral alignment / Telemetry-consistent ATT&CK interpretation; Not direct signature classification.

Epistemic Attribution Disclaimer: The engine correlates passive wire flow features with ATT&CK techniques based on statistical telemetry distributions. It does NOT perform deep packet payload inspection, exploit signature matching, or cryptographic payload verification. Mappings provide contextual guidance for threat hunting, not forensic proof of compromise.

T1046: Network Service Discovery

T+1 (60s)

TACTIC: Discovery

Behavior consistent with automated port scan and network service enumeration.

T1190: Exploit Public-Facing Application

T+3 (180s)

TACTIC: Initial Access

Downstream transition probability indicates heightened potential for service exploitation attempts.

T1498: Network Denial of Service

T+5 (300s)

TACTIC: Impact

Volumetric packet burst consistent with network flooding or resource exhaustion.

EXPLAINABILITY & CAUSAL EVIDENCE

Feature Influence & Evidence Chain

Counterfactual perturbation drivers and corroborating forensic evidence nodes.

PRIMARY FEATURE ATTRIBUTION DRIVERS

mean_iat

Inter-arrival transmission gap collapses into high-velocity automated burst pacing.

unique_dst_ports

Target port cardinality surges across endpoints, indicating systematic port scanning.

mean_swin

Client TCP window size shrinks, characteristic of scripted flooding tools.

delta_flow_count

New 5-tuple flow initiation velocity accelerates significantly above stationary baseline.

SUPPORTING VS CONTRADICTIONARY TELEMETRY

unique_dst_ports

SUPPORTING

Destination port cardinality surged by +700%, indicating active network reconnaissance.

mean_iat

SUPPORTING

Inter-arrival timing collapsed by -73.6%, reflecting automated high-speed request pacing.

tcp_rst_count

CONTRADICTIONARY

TCP connection abort rate remains low and stable, counterbalancing brute-force hypotheses.

Scientific Limitations & Governance

Strictly verified boundaries and methodological constraints.

- Model Activation Scores:** Forward-step scores represent raw LSTM neural state activations and transition signals, not calibrated actuarial event probabilities.
- Contextual MITRE Interpretation:** All MITRE ATT&CK mappings are contextual behavioral alignments derived from telemetry distributions, not direct ground-truth signature classifications.
- Horizon Confidence Decay:** Simulation certainty naturally diminishes as the rollout deepens from T+1 (+60s) to T+5 (+300s) due to recursive latent state variance.
- RTT Exclusion Integrity:** Round-trip time metrics are deliberately withheld from passive wire captures to prevent synthetic data imputation and preserve scientific provenance.

Technical Specification & Reproducibility Audit

Deterministic system environment, model configuration, and pipeline provenance.

MODEL ARCHITECTURE LSTM World Model Recursive latent dynamic state rollout	CANONICAL STATE 45-Feature Contract Layer 3/4 passive telemetry vector	TEMPORAL RESOLUTION 60-Second Windows Discrete tumbling slices (zero overlap)
LOOKAHEAD HORIZON T+1 → T+5 (+300s) Multi-step forward projection	REFERENCE BASELINE Persistence Baseline Empirical zero-drift standard	INPUT TELEMETRY PARQUET Passive Capture 2 windows · 120s
PROVENANCE & AUDIT VERIFIED REFERENCE Deterministic Pipeline v1.0	EPISTEMIC SAFETY GATE EARLY STAGE (<8 Windows) Precondition gate enforced	